

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|--------------------------------------|---|
| 1. Pivot B | (A) $A = LU$: a lower-triangular L times an upper-triangular U . |
| 2. Multiplier D | (B) The first nonzero entry in a row, used to clear the entries below it. |
| 3. Upper triangular (U) G | (C) A matrix with no inverse — its determinant $ad - bc$ is 0. |
| 4. Inverse matrix H | (D) $\ell = (\text{entry to clear}) \div (\text{pivot})$; how many pivot rows you subtract. |
| 5. Singular matrix C | (E) The identity with two rows swapped; it records a row exchange. |
| 6. LU factorization A | (F) The matrix A^T formed by flipping A across its diagonal (rows $\leftrightarrow$ columns). |
| 7. Permutation matrix E | (G) A matrix with all zeros below the main diagonal. |
| 8. Transpose F | (H) A^{-1} , the matrix that undoes A : $A^{-1}A = I$. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- To clear the entry below a pivot p , you subtract ℓ times the pivot row, where ℓ is (A) $p \times (\text{entry})$ (B) $(\text{entry}) \div p$ **C** (C) $p \div (\text{entry})$ (D) $p - (\text{entry})$
- In the factorization $A = LU$, the matrix L holds (A) the pivots (B) the solution $\mathbf{x}$ **(C) the multipliers, with 1's on the diagonal** **C** (D) all zeros
- A 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is **singular** (no inverse) exactly when (A) $ad - bc = 0$ **C** (B) $ad - bc = 1$ (C) $a = d$ (D) it is upper triangular
- If P is a permutation matrix, the product PA (A) scales the rows of A **(B) swaps rows of A** **C** (C) transposes A (D) inverts A
- For any two matrices that can be multiplied, $(AB)^T =$ (A) $A^T B^T$ (B) AB (C) BA **(D) $B^T A^T$** **C**
- Two equations in two unknowns have **infinitely many** solutions when the two lines (A) cross once (B) are parallel but different **(C) are the same line** **C** (D) are perpendicular

Part C — Short Answer & Computation (30 pts)

Show your work.

1. Solve the system by **elimination**, then back-substitute and check. State the pivot and the multiplier you used.

$$x + 2y = 4, \quad 3x + 4y = 10.$$

$$\begin{aligned} \text{pivot} &= 1, \quad \ell = 3 \div 1 = 3 \\ (4 - 6)y &= 10 - 12 \Rightarrow -2y = -2, \quad y = 1 \\ x + 2(1) &= 4 \Rightarrow x = 2 \\ 3(2) + 4(1) &= 10 \quad \checkmark \end{aligned}$$

2. Row-reduce $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 4 & 10 & 8 \end{bmatrix}$ to upper-triangular U . List the three multipliers and the pivots.

$$\begin{aligned} R_2 - 2R_1, \quad R_3 - 4R_1 &\Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 4 \end{bmatrix} \\ R_3 - 2R_2 &\Rightarrow U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \\ \ell_{21} &= 2, \quad \ell_{31} = 4, \quad \ell_{32} = 2, \quad \text{pivots } 1, 1, 2 \end{aligned}$$

3. Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$. Use the $ad - bc$ formula to find A^{-1} , then check that $A^{-1}A = I$.

$$\begin{aligned} ad - bc &= 2 \cdot 2 - 1 \cdot 3 = 1 \\ A^{-1} &= \frac{1}{1} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \\ A^{-1}A &= \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \checkmark \end{aligned}$$

4. Using the inverse from the previous problem, solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ (compute $\mathbf{x} = A^{-1}\mathbf{b}$).

$$\begin{aligned} \mathbf{x} &= \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 - 4 \\ -9 + 8 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\ \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} &= \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad \checkmark \end{aligned}$$

5. Factor $A = \begin{bmatrix} 2 & 4 \\ 6 & 7 \end{bmatrix}$ as $A = LU$. Write L and U , then verify $LU = A$.

$$\begin{aligned} \text{pivot} &= 2, \quad \ell = 6 \div 2 = 3 \\ U &= \begin{bmatrix} 2 & 4 \\ 0 & -5 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \\ LU &= \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 7 \end{bmatrix} = A \quad \checkmark \end{aligned}$$

6. Using the L and U from the previous problem, solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ in **two sweeps**: first $L\mathbf{c} = \mathbf{b}$ (forward), then $U\mathbf{x} = \mathbf{c}$ (back).

$$\begin{aligned} L\mathbf{c} = \mathbf{b} : \quad c_1 &= 2, \quad 3c_1 + c_2 = 1 \Rightarrow c_2 = -5 \\ \mathbf{c} &= \begin{bmatrix} 2 \\ -5 \end{bmatrix} \\ U\mathbf{x} = \mathbf{c} : \quad -5x_2 &= -5 \Rightarrow x_2 = 1, \quad 2x_1 + 4 = 2 \Rightarrow x_1 = -1 \\ \mathbf{x} &= \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \checkmark \end{aligned}$$

7. (a) The matrix $A = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$ has a 0 in the top-left pivot spot. Write the permutation matrix P that swaps the two rows, and give PA . (b) Write the transpose of $B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ and state whether B is symmetric.

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad PA = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = B \Rightarrow \text{symmetric}$$

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. (a) Compute $ad - bc$. Does A have an inverse? What does that say about solving $A\mathbf{x} = \mathbf{b}$? (b) How many solutions does $A\mathbf{x} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ have? (c) How many does $A\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ have? Justify (b) and (c) using elimination or the row/line picture.

$$ad - bc = 1 \cdot 4 - 2 \cdot 2 = 0 \Rightarrow \text{singular, no inverse}$$

$$(b) \quad R_2 - 2R_1 : 6 - 2(3) = 0 \Rightarrow 0 = 0, \text{ infinitely many}$$

$$(c) \quad R_2 - 2R_1 : 1 - 2(1) = -1 \Rightarrow 0 = -1, \text{ no solution}$$

So $A\mathbf{x} = \mathbf{b}$ is never uniquely solvable: (b) is the same line, (c) parallel lines.

2. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. (a) Compute $A^T A$ and check that it is symmetric. (b) Explain why $A^T A$ is *always* symmetric for any matrix A , using the reversal rule $(AB)^T = B^T A^T$.

$$A^T A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$$

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

(a) The off-diagonal entries match ($5 = 5$), so $A^T A$ is symmetric. (b) It equals its own transpose by the reversal rule, so it always is.